CASEY CRANE

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EDUCATION

BS 2015, MS 2020, PhD 2026, Mathematics, University of Texas at Dallas

PUBLICATIONS

Dissertation

- "Consensus-breaking global Hopf bifurcation in memory-based multi-agent systems." PhD Dissertation. Advisor: Dr. Wieslaw Krawcewicz.

Journal Publications

- Chaoquan Chen, **Casey Crane**, and Travis Hensley. "Global Hopf bifurcation in symmetric configurations of distributed delay differential equations" *Communications in Pure and Applied Analysis* 27 (2026): 170–194.
- Ziad Ghanem, **Casey Crane**, and Jingzhou Liu. "Global bifurcation of non-radial solutions for symmetric sub-linear elliptic systems on the planar unit disc" *Journal of Fixed Point Theory and Applications* 26 (2024): 44.
- Yameng Duan, **Casey Crane**, Wieslaw Krawcewicz, and Huafeng Xiao. "Periodic solutions in reversible symmetric second order systems with multiple distributed delays" *Journal of Differential Equations* 401 (2024): 282–307.

Preprints

- **Casey Crane**. "Global Hopf bifurcation and symmetric periodic solutions in multi-agent systems with neutral distributed delays" arXiv:2604.20740 (2026). In revision at SIADS.
- **Casey Crane**. "Trust as hysteresis: analytically tractable strategies for the iterated prisoner's dilemma" Preprint (2026). Submitted to J. Theor. Biol.
- **Casey Crane** and Ziad Ghanem. "Unbounded branches of non-radial solutions to semilinear elliptic systems on a disc and their patterns" arXiv:2504.06519 (2025). In revision at TMNA.
- **Casey Crane** and Ziad Ghanem. "Unbounded branches of non-radial solutions to semilinear elliptic systems on a unit ball in $\mathbb{R}^3$ and their patterns" arXiv:2512.14957 (2025). Preprint.

RESEARCH SUMMARY

My research uses equivariant degree theory to study symmetry-breaking bifurcation and global continuation in delay equations and highly symmetric networks of interacting agents. Building on the study of consensus protocols in multi-agent systems, I ask what happens after consensus breaks down: I use topological methods to prove the existence and global continuation of branches of symmetric periodic solutions, and to classify them by their spatio-temporal symmetry. These are periodic multiconsensus states, in which the agent population fragments into clusters that oscillate out of phase. This work also led me to identify a class of delay equations I call pseudoneutral: equations written in the implicit form of neutral equations that are in fact of retarded type, arising as the mean-field limit of populations with distributed neutral delays. I pair these abstract results with targeted numerical analysis, and I develop applications in fields—networked control, economics, collective behavior—where high symmetry obstructs classical bifurcation theory and where these methods remain underutilized. Recent work includes UAV formation control and networks of coupled financial markets.

TEACHING EXPERIENCE

Teaching Associate, University of Texas at Dallas *Spring 2026*
Instructor of record: Differential Calculus.

Teaching Assistant, University of Texas at Dallas *Fall 2021 – Fall 2025*
Courses supported: Calculus I–III (single variable, vector, and multivariable calculus), Differential Equations, Functional Analysis (graduate).

TALKS

- *Invited.* "Global symmetric Hopf bifurcation using equivariant degree." Hunan Normal University, Changsha, China, 2023.
- *Contributed.* "Global Hopf bifurcation in symmetric configurations of distributed delay differential equations." SIAM Texas-Louisiana Section Annual Meeting, Baylor University, 2024.